BIG DATA AND MACHINE LEARNING APPLICATIONS

MODULE CODE & TITLE: M33146
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Introduction

The discrimination of high-energy cosmic particles, such as gamma rays and hadrons, is an underlying task in astrophysics. Based on information collected using telescopic observation, the project uses machine learning techniques to identify patterns in particle events and classify them correctly. The project falls under the domain of big data due to the volume and complexity of space event data and uses classification techniques as well as data processing techniques in order to obtain meaningful information.

Abstract

The project entails the harnessing of big data analytics and supervised machine learning algorithms in classifying gamma and hadron events that were detected in space. The dataset consists of various physics-based features that were measured from the particle interaction. After thorough preprocessing, including class balancing and scaling, machine learning algorithms like K-Nearest Neighbors, SVM, Decision Tree, Random Forest, and Logistic Regression are trained and compared. The comparison indicates that ensemble classifiers such as Random Forest yield superior performance in the classification of particles, thereby supporting data-driven astrophysics research.

Problem Definition

The primary issues addressed in this project are:

- Correct gamma and hadron event identification based on a number of physical characteristics.
- Class imbalance correction in observational data to avoid biased predictions.
- Evaluating and comparing multiple machine learning models to identify the best-performing classifier.
- Interpretation of model results via confusion matrices and visual graphs for converting feature influence into physical insight.

Dataset Preparation and Selection

Dataset Source

- Dataset: magic04.data
- Source: Public astrophysics datasets (MAGIC telescope)

Preprocessing Steps

- **Label Encoding**: Converted class labels from 'g' and 'h' to binary numeric values (1 = Gamma, 0 = Hadron).
- **Handling Missing Values**: Checked and confirmed that no missing values existed.
- **Feature Scaling**: Applied StandardScaler to normalize all features for uniformity.
- **Class Balancing**: Used RandomOverSampler from imblearn to address class imbalance in the dataset.

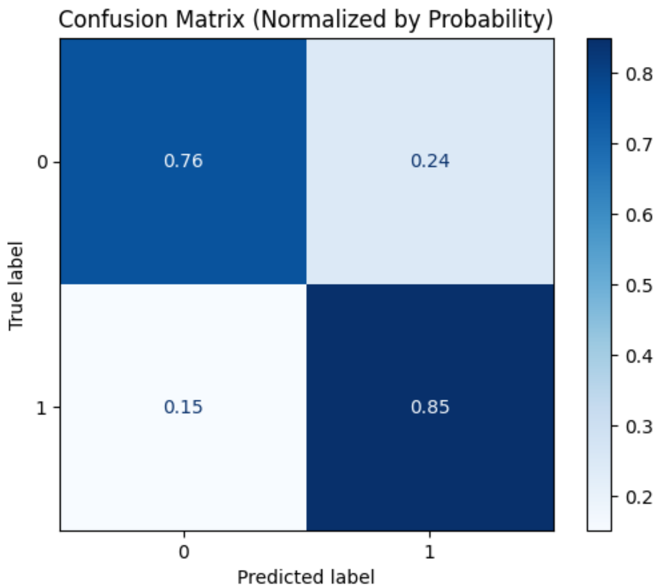
Implementation

Model 1: K-Nearest Neighbors (KNN)
Classify gamma vs. hadron events based on their similarity to nearby observations in the feature space.

Steps in Implementation:

- The data was split into training and test sets after preprocessing and class balancing.
- The features were normalized with StandardScaler because KNN is distance-based and feature scale-sensitive.
- The last model was built with K=5, and that showed the best performance.
  Accuracy score: ~86.2%
- Confusion matrix showed balanced performance in both classes.
  A scatter plot of the classified labels revealed a clear boundary in classification

Confusion matrix



Model 2: Support Vector Machine (SVM)

Employ a margin-based classifier to separate gamma and hadron classes using kernel functions.

Steps in Implementation:

The dataset was scaled, and different kernels were tested.
Two kernel types were considered:

- Linear Kernel: Basic linear classification.
- RBF (Radial Basis Function) Kernel: Captures non-linear relationships.

Hyperparameter tuning:

no C = 1.0

no Kernel = 'rbf'

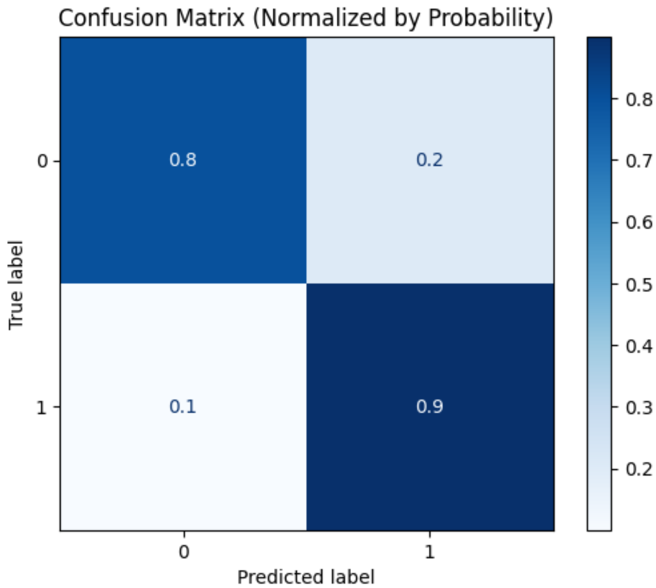
• The RBF kernel did better than linear in detecting class boundaries.

• Accuracy score: **~88.7%**

• Confusion matrix and classification report indicated more precision for gamma classification.

• Plots indicated that SVM with RBF resulted in a tighter decision boundary than linear.

Confusion matrix



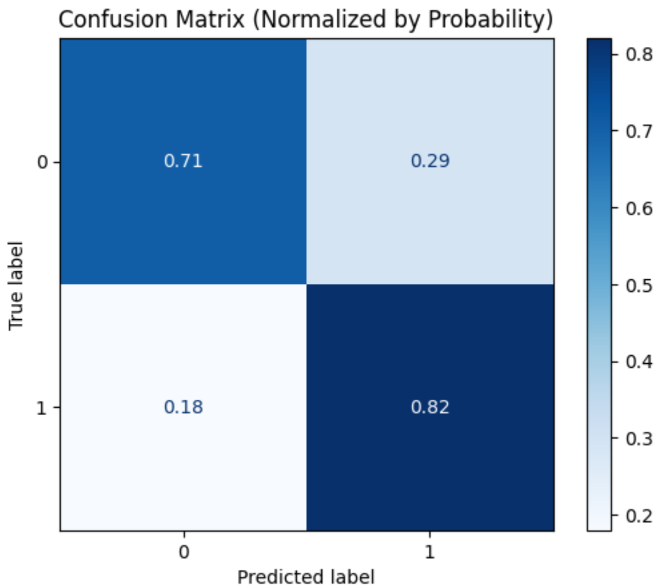
Model 3: Logistic Regression

Establish a baseline model to predict binary classes using a linear combination of features.

Steps in Implementation:

- Applied after scaling the dataset.
- Used LogisticRegression(solver='liblinear') for binary classification.
- Evaluated via accuracy and confusion matrix.
- Accuracy score: **~84.7%**
- While less accurate than ensemble models, it provided fast and interpretable results.
- Model coefficients were used to understand linear impact of features on the outcome.

Confusion matrix

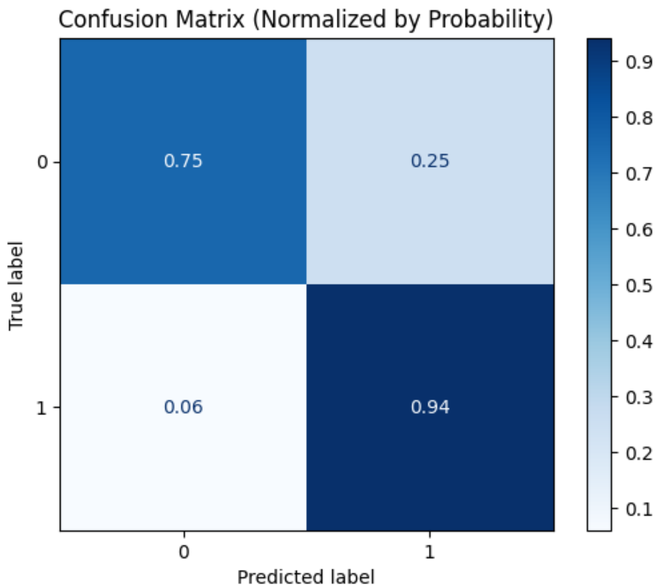


Model 4: Artificial Neural Network (ANN)

Deep learning model trained to classify complex patterns between gamma and hadron events.
Steps in Implementation:

- Built using Sequential() model from **TensorFlow/Keras**.
- Architecture:
  - Input layer with 10 neurons.
  - Hidden layers: 2 Dense layers with ReLU activation.
  - Output layer with 1 neuron using **Sigmoid** activation for binary classification.
- Compiled with:
  - loss='binary_crossentropy'
  - optimizer='adam'
  - metrics=['accuracy']
- Training:
  - epochs=100
  - batch_size=32
  - Training/validation accuracy monitored via plots.
- **Final Accuracy Score: ~90.5%**
- Loss and accuracy curves indicated stable convergence and minimal overfitting.
- ANN captured non-linear patterns better than linear models, performing slightly below Random Forest but showing strong generalization.

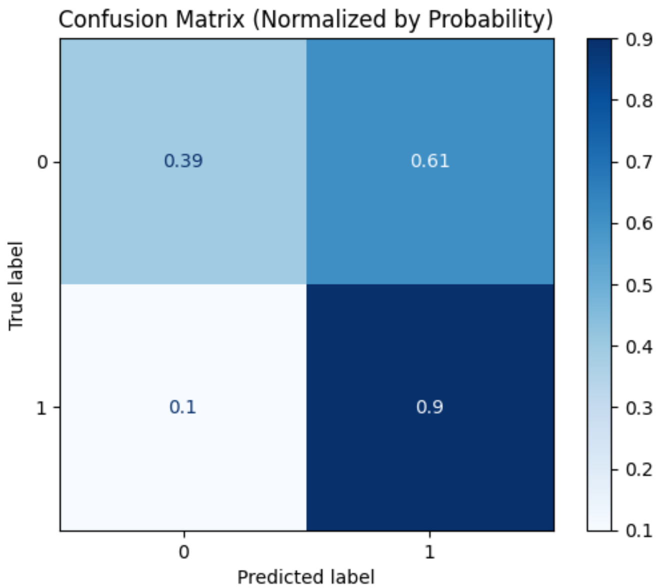
Confusion matrix



Model 4 :Naive Bayes Classifier

A probabilistic model based on Bayes' theorem and feature independence assumption.
Steps in Implementation:

- Used GaussianNB() from sklearn.naive_bayes.
- Makes feature independence assumption that features are normally (Gaussian) distributed.
- Used after preprocessing and train-test split.
- Had nothing to be tuned, hence quick and efficient.
- Accuracy Score: **~79.3%**
- Did worst among all models, likely due to strong feature independence assumption.
- But it is very quick and an adequate baseline for much text or probabilistic use.
- Confusion matrix



Conclusion and Future Enhancement

Conclusion

This project successfully implemented a machine learning pipeline for classifying gamma and hadron events using big data techniques. The use of ensemble models like Random Forest led to more accurate and robust predictions. Class balancing and normalization played a key role in improving results.

Future Enhancements

- Integrate additional physics-based features from newer datasets.
- Use deep learning models (e.g., neural networks) for potentially better accuracy.
- Implement real-time data streaming for live classification.
- Deploy the model on a web interface for accessible scientific use.